

Multiple Choice (10 marks)

1. Which of the following sets of bit strings CANNOT be described with a regular expression?
 - (a) All bit strings whose number of zeros is a multiple of five
 - (b) All bit strings starting with a zero and ending with a one
 - (c) All bit strings with an even number of zeros
 - (d) All bit strings with more ones than zeros
2. Which of the following is NOT a reasonable justification for choosing to busy-wait on an asynchronous event?
 - (a) The wait is expected to be short.
 - (b) A busy-wait loop is easier to code than an interrupt handler.
 - (c) There is no other work for the processor to do.
 - (d) The program executes on a time-sharing system.
3. Of the following, which gives the best upper bound for the value of $f(N)$ where f is a solution to the recurrence $f(N) = f(N/2) + f(N/3) + 1$?
 - (a) $O(\log N)$
 - (b) $O(N \log N)$
 - (c) $O(\log N) + O(1)$
 - (d) $O((\log N)^2)$
4. In the Internet Protocol (IP) suite of protocols, which of the following best describes the purpose of the Address Resolution Protocol (ARP)?
 - (a) To translate Web addresses to host names
 - (b) To determine the IP address of a given host name
 - (c) To determine the hardware address of a given host name
 - (d) To determine the hardware address of a given IP address
5. Which of the following is true of interrupts?
 - (a) They are generated when memory cycles are "stolen".
 - (b) They are used in place of data channels.
 - (c) They can indicate completion of an I/O operation.
 - (d) They cannot be generated by arithmetic operations.

True / False (10 marks)

Answer True or False

6. True or False: The number of distinct functions mapping a finite set A with m elements into a finite set B with n elements is m^n .
7. True or False: Reference counting requires additional space overhead for each memory cell to maintain a count of references to it.
8. True or False: The symbol table is a data structure in a compiler that manages information about the identifiers in the source program.
9. True or False: Finding a shortest cycle in an undirected graph G can be solved in polynomial time.
10. True or False: A heap area is usually not represented in a subroutine's activation record frame.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find numbers divisible by m or n from a list of numbers using lambda function.
12. Write a function to check whether the given number is undulating or not.

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